setting. However, this analysis is very distinct from our work, not only because it places itself in the kernel setting with Gaussian design, but also because it assumes the distillation setting, where the "generation" stage is carefully optimized for the next stage training. In the case of synthesized data in the wild, this assumption can of course not be made.

Emergence of "Skills" and Scaling Laws: Scaling laws give us an insight on bang-for-the-buck style trade-off for model training. However, cross-entropy loss is not a goal in and of itself: we want to train models that are endowed with a larger and larger skill set as we scale them up. For instance, Gordon et al. (2021) provide intuition and empirics for the scaling of BLEU score for MT with cross-entropy loss as

$$BLEU(L) \approx Ce^{-kL},$$

demonstrating "emergence" of good BLEU performance with scale. This type of "emergence" has been massively confirmed in (Wei et al., 2022), where a working definition of "emerging" is "not present in smaller models, but appears in larger models". In this sense, Wei et al. (2022) demonstrate empirically a large number of "skills" appearing with scale, like Multi-Task NLU, Modular arithmetic, word unscrambling and transliteration.

A theoretical model, providing an underpinning of the necessity of scaling laws for the emergence of skill has recently been given by (Arora & Goyal, 2023). They analyse "emergence" with the scaling laws as a departure point in a model that links cross-entropy loss in LLMs to basic skills to show that scaling laws enable the model to learn (and generalize) efficiently.

Strengthening the tie between scaling laws and emergent skill, albeit in the *opposite* direction, Michaud et al. (2023) posit that skills that emerge in "quanta" imply a scaling law of the loss. Related, Chen et al. (2023) assume a hierarchy of skills to derive data curation mechanisms to precipitate the emergence of skills, though they do not allude to scaling laws directly.

B. Complimentary Figures for Sections 2, 3 and 4

Hutter LLM. Figures 8, 9 and 10 further illustrate our theory for simple Hutter LLM.

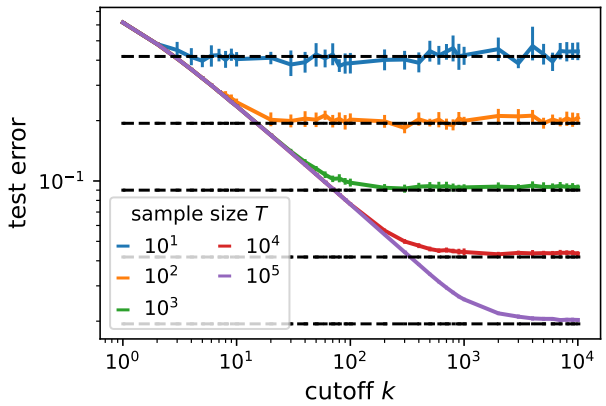

Figure 8. **Scaling on Hutter LLM for Varying T.** Empirical confirmation of Theorem 2.1. Here, $\beta = 3/2$ and error bars correspond to 10 iid runs of sampling AI-generated data (i.e the distribution q). Broken lines correspond to the Hutter rate $T^{-(\beta-1)/\beta}$, for varying k and different values of T. Figure 3, left, illustrates the same for varying T and several settings of k. Note the perfect match with the theorem.

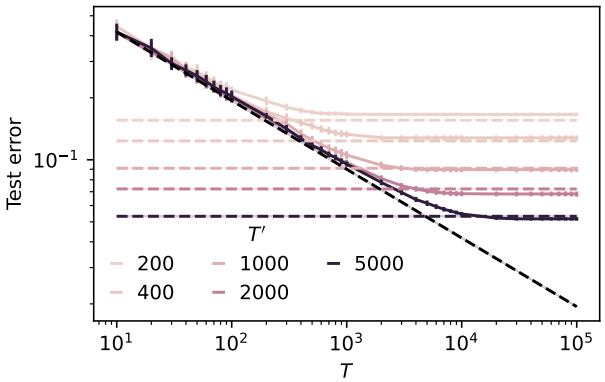

Figure 9. **Scaling on Hutter LLM for Varying k.** A sample of size T_0 is used to approximate the true distribution p via p_{AI}. Then, a Hutter-type model is learned on a sample of size T from p_{AI}, and evaluated on the true data distribution p. Each horizontal line corresponds to the asymptote $k^{-\beta c} \asymp T_0^{-c}$, for different values of T_0. The diagonal line corresponds to T^{-c}.

Hutter++. We now provide complementary illustrations of predictions made from the theory we have developed for the generalized Hutter models as in Equation (16) in Section 4, without departing from our theoretical assumptions. We also show how theory from the infinite memory model in Section 2 continues to hold in this bigram setting. Figure 11 confirms the scaling law of Theorem 4.2.

In Figure 3 (middle) we have seen an illustration of the translated scaling curves under n-fold synthesized data in the Hutter LLM. Figure 12 illustrates this phenomenon for the slightly more complex tailed bigram model.

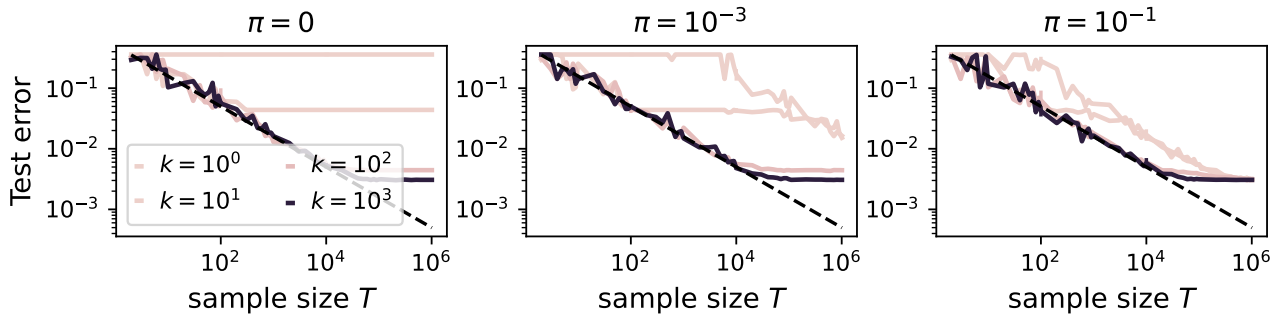

Figure 10. Empirical Validation of Theorem 3.2. The broken line corresponds to the $T^{-(1-1/\beta)}$ scaling law that would hold throughout in the absence of pollution. Notice the grokking behavior predicted by the theorem. For this experiment, the Zipf exponent of the true data distribution p is $\beta = 2$.

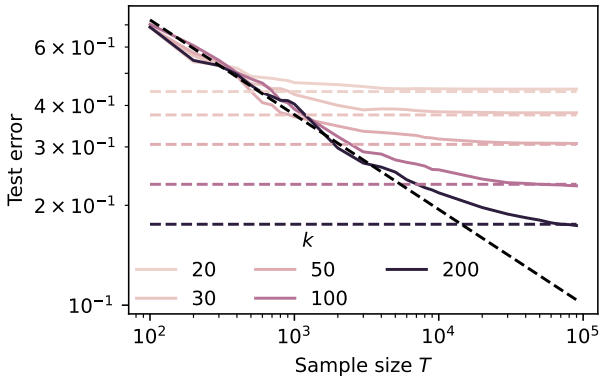

Figure 11. **Model Collapse for Hutter++.** Empirical confirmation of Theorem 4.2. Here $p(j \mid i)$ is as in (16), with $\beta = 7/5$. The horizontal broken lines correspond to $k^{-\beta c}$ for different values of k, where $c := \min(1 - 1/\beta, 1/2)$. The diagonal broken line corresponds to T^{-c} (classical error rate without cutoff).

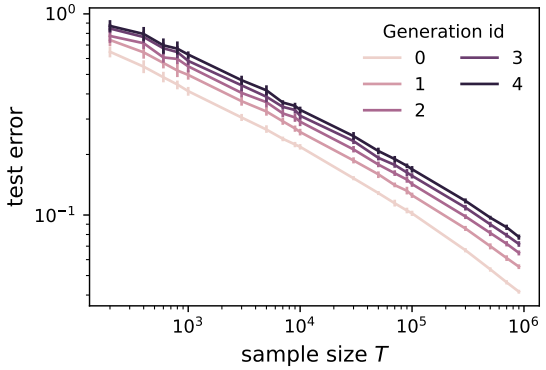

Figure 12. **Hutter++ Model on Paired Bigram Data.** Empirical confirmation of Theorem 2.4 for probabilistic paired bigram data with $\beta = 3/2$, $T_0 = T$ across all generations with no additional tail-cutting, regeneration for 9 times. The result verifies the model collapse across generation.

Both Figures 3 (middle) and 12 illustrate the setting where each model consumes as much training data as its predecessor ($T_0 = T$). We now relax the assumption that each successive model has strictly the same amount of training data as its predecessor. We assume that the generation 0 model is trained on T_0 (here, $T_0 = 100,000$) amount of original data to generate AI data for generation 1. All future generations, starting from generation 2, are trained on data generated by the most powerful model from the previous generation ($T = 1,000,000$ data in this case). Figure 13 (for Hutter LLM) and 14 (for Hutter++ on paired bigram data) show the resulting scaling behavior. We take this setting even further by adding a top-p^{inf} tail cutting mechanism and a temperature scaling mechanism for each synthetic data generation. Figure 6 cuts at $p = 0.95$ and Figure 16 at temperature 0.9.

We now study mixing of clean and synthesized data in the bigram setting. Figures 17 and 18 add top-p^{inf} tail-cutting when synthesizing, and start with $T_0 = 10,000$ original data samples, which are successively blended with synthesized data from the largest model. Note that in this setting we observe a reversion of scaling laws with increased AI data. This needs to be compared with the orange curve in Figure 20 in the deterministic Hutter setting. The probabilistic nature of the bigram models leads to a new effect here.

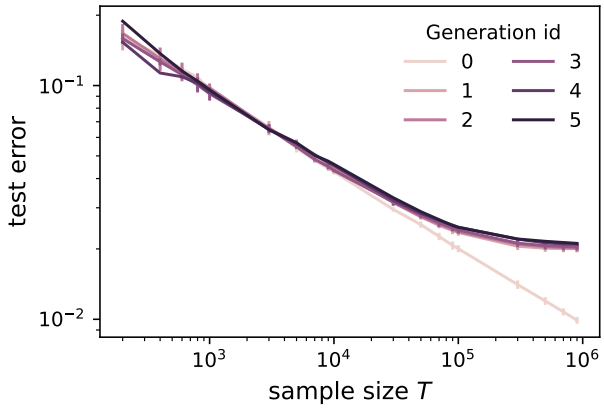

Figure 13. **Empirical Hutter LLM.** Bigram model with deterministic labeling function. Initial model trained on $T_0 = 100,000$ samples. It generates T samples for Gen 1. Starting from Gen 2 models are trained on data generated by the most powerful model from the previous generation. $\beta = 3/2$. In this setting, there is mild model collapse coming from the finite sample bias.

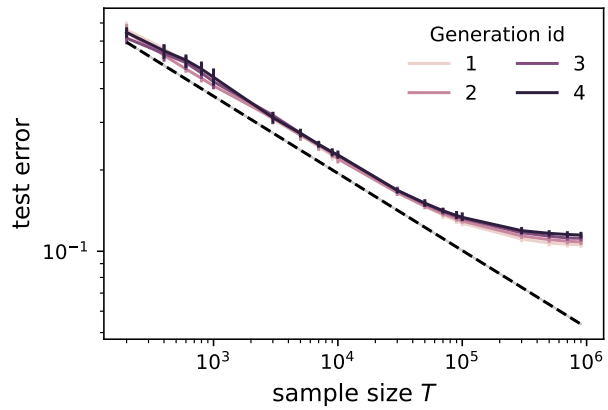

Figure 14. **Empirical Hutter++ Model.** Same setting as in Figure 6. Initial model trained on $T_0 = 100,000$ samples. No top-p^{inf} inference or temperature scaling is used. $\beta = 3/2$. In this setting, there is mild model collapse coming from the finite sample bias as well.

C. Proofs for the infinite memory (Hutter) Model (Sections 2 and 3)

C.1. Proof of Theorem 2.1

Observe that the model $\widehat{f}$ makes an error on i if and only if the ith "skill" never occurred in the training dataset $\mathcal{D}_T$, i.e either (1) $i \geq k+1$, or (2) $1 \leq i \leq k$ and $i_t \neq i$ for all $t \in [T]$. We deduce that

$$\begin{aligned} E_{test} = \mathbb{P}_{i\sim p}(\widehat{f}(i) \neq y_i) &= \sum_{i\geq k+1} p_i + \sum_{1\leq i\leq k} p_i(1-p_i)^T \\ &\asymp k^{-(\beta-1)} + \sum_{1\leq i\leq k} p_i e^{-p_i T}, \end{aligned}$$

where $c := 1 - 1/\beta \in (0,1)$, and we have used the elementary fact that $\sum_{i\geq k+1} i^{-\beta} \asymp k^{-(\beta-1)}$ for large k. For the second sum, we will need the following lemma.

Lemma C.1. *The following identity holds*

$$T^c \sum_{i=1}^{k} p_i e^{-Tp_i} \asymp \Gamma(c, Tk^{-\beta}) - \Gamma(c,T) = O(1), \tag{22}$$

where $\Gamma(s,x) := \int_x^\infty u^{s-1}e^{-u}\mathrm{d}u$ *defines the incomplete gamma function. In particular, for* $k=\infty$ *and large* T*, it holds that* $\sum_{i=1}^\infty p_i e^{-Tp_i} \asymp T^{-c}$.

Proof. Consider the function $h(z) := ze^{-Tz}$ for $z \in (0,1)$. Its derivative is $h'(z) = e^{-Tz}(1-Tz)$. Thus, h is increasing on $(0, 1/T)$ and decreasing on $(1/T, \infty)$. Furthermore, note that $p_i \leq 1/T$ iff $i \geq T^{1/\beta}$. We deduce that

$$\sum_{i=1}^{k} p_i e^{-Tp_i} \asymp \int_1^k x^{-\beta} e^{-Tx^{-\beta}} \mathrm{d}x.$$

Under the change of variable $u = u(x) := Tx^{-\beta}$, we have $x = x(u) = (u/T)^{-1/\beta}$ and so $\mathrm{d}x = -(T^{1/\beta}u^{-1-1/\beta}/\beta)\mathrm{d}u$.